

Project Design Phase
Problem – Solution Fit Template

Date	09/03/2026
Team ID	SWTID-2026-4275
Project Name	SpendWise
Maximum Marks	2 Marks

Problem–Solution Fit Overview:

The Problem–Solution Fit ensures that beginners learning backend development can clearly understand key concepts through a practical approach. SpendWise addresses this by providing a simple project that demonstrates API development, JWT authentication, and MongoDB integration in an easy, hands-on way.

Purpose:

- Help beginners understand how a backend project is structured and how APIs work.
- Provide a simple project to practice CRUD operations.
- Demonstrate how JWT authentication and protected routes function.
- Show how to connect Node.js and Express with MongoDB.
- Provide a minimal backend project that can be easily tested using Postman or Thunder Client.

Problem Statement:

Many beginners find backend development difficult because learning resources are often either too basic or overly complex. As a result, it becomes hard to understand how important components like authentication, database connections, and CRUD operations work together in a real application. This creates confusion when building structured backend systems and limits practical understanding of REST APIs and JWT-based security.

Solution:

SpendWise is a **simple Node.js, Express, and MongoDB backend project** designed for learning purposes.

- It provides **User Registration and Login** with **JWT authentication**.
- After login, users can perform **CRUD operations** (Create, Read, Update, Delete) on their own data.
- The project uses a **clean folder structure** (controllers, models, routes, middleware) to teach best practices.
- APIs can be easily tested using **Postman or Thunder Client**, making it beginner-friendly and practical.